Finely Stratified Rerandomization Designs

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Inference. Asymptotically exact inference on superpopulation parameters, efficient conservative inference on finite population counterparts.

Related Literature

Stratified Randomization. Bugni et al. (2018), Bai et al. (2022), Bai (2022), Cytrynbaum (2024), Bai et al. (2024).

Rerandomization. Morgan and Rubin (2012), Li et al. (2018), Li and Ding (2020), Wang et al. (2021), Wang and Li (2022), Wang and Li (2024).

Nonlinear/Minimax Designs. Ding and Zhao (2024), Schindl and Branson (2024), Liu et al. (2023).

Finely Stratified Rerandomization

Consider $W_{1:n} = (Y_i(0), Y_i(1), \psi_i, h_i)_{i=1}^n \overset{\text{iid}}{\sim} F$, where $\psi \in \mathbb{R}^{d_\psi}$ are stratification variables and $h \in \mathbb{R}^{d_h}$ are rerandomization variables.

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Stratified Rerandomization.

1. Match units into groups $|s| = k$ satisfying

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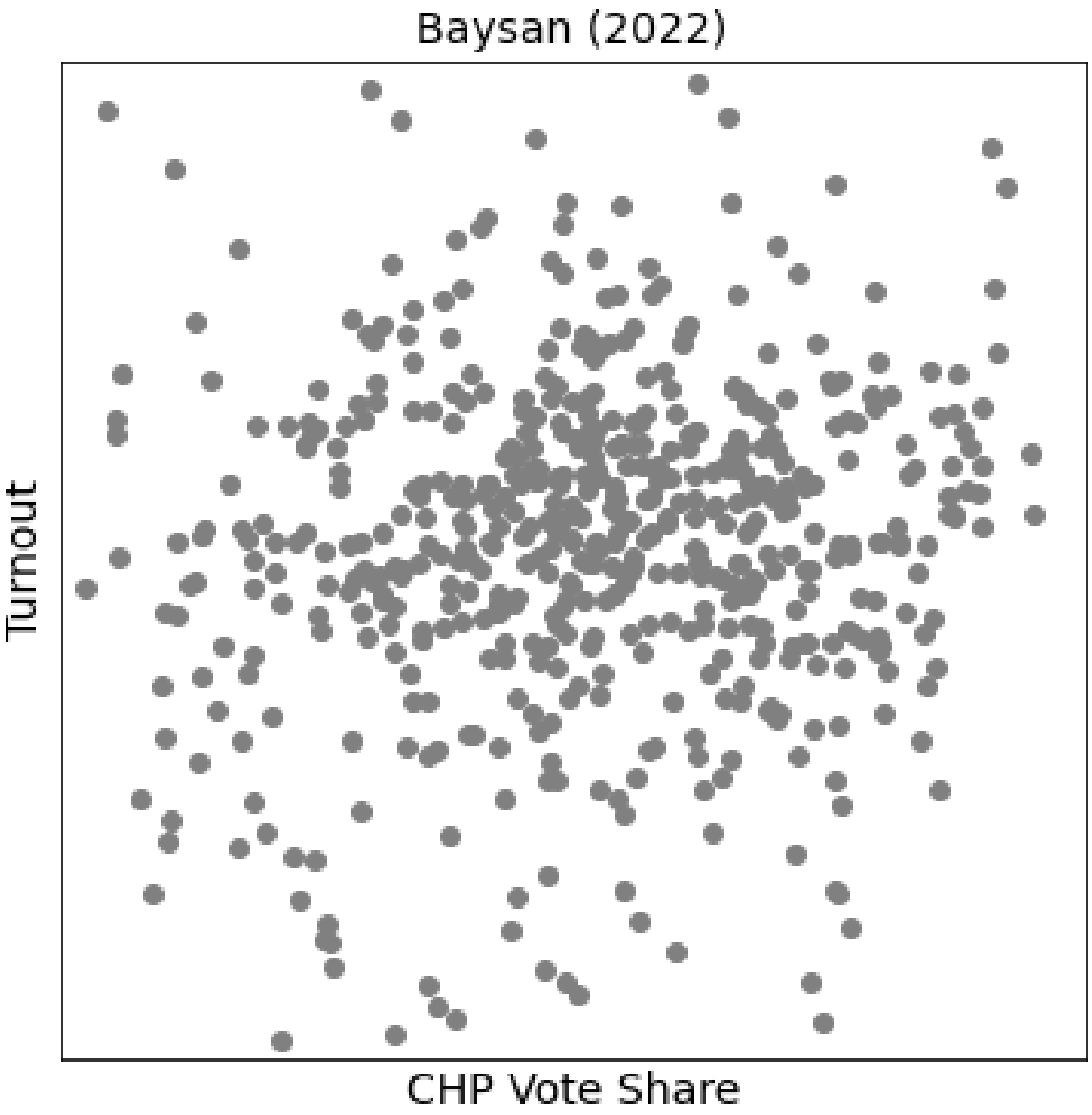
2. Randomize $p = a/k$ units to $D_i = 1$ in each group s.
3. For **acceptance region** $A \subseteq \mathbb{R}^{d_h}$, accept $D_{1:n}$ iff

$$\sqrt{n}(\bar{h}_1 - \bar{h}_0) \in A, \quad \bar{h}_d = \frac{\sum_i h_i \mathbb{1}(D_i = d)}{\sum_i \mathbb{1}(D_i = d)}$$

E.g. $A = B(0, \epsilon)$, check $|\sqrt{n}(\bar{h}_1 - \bar{h}_0)| \leq \epsilon$.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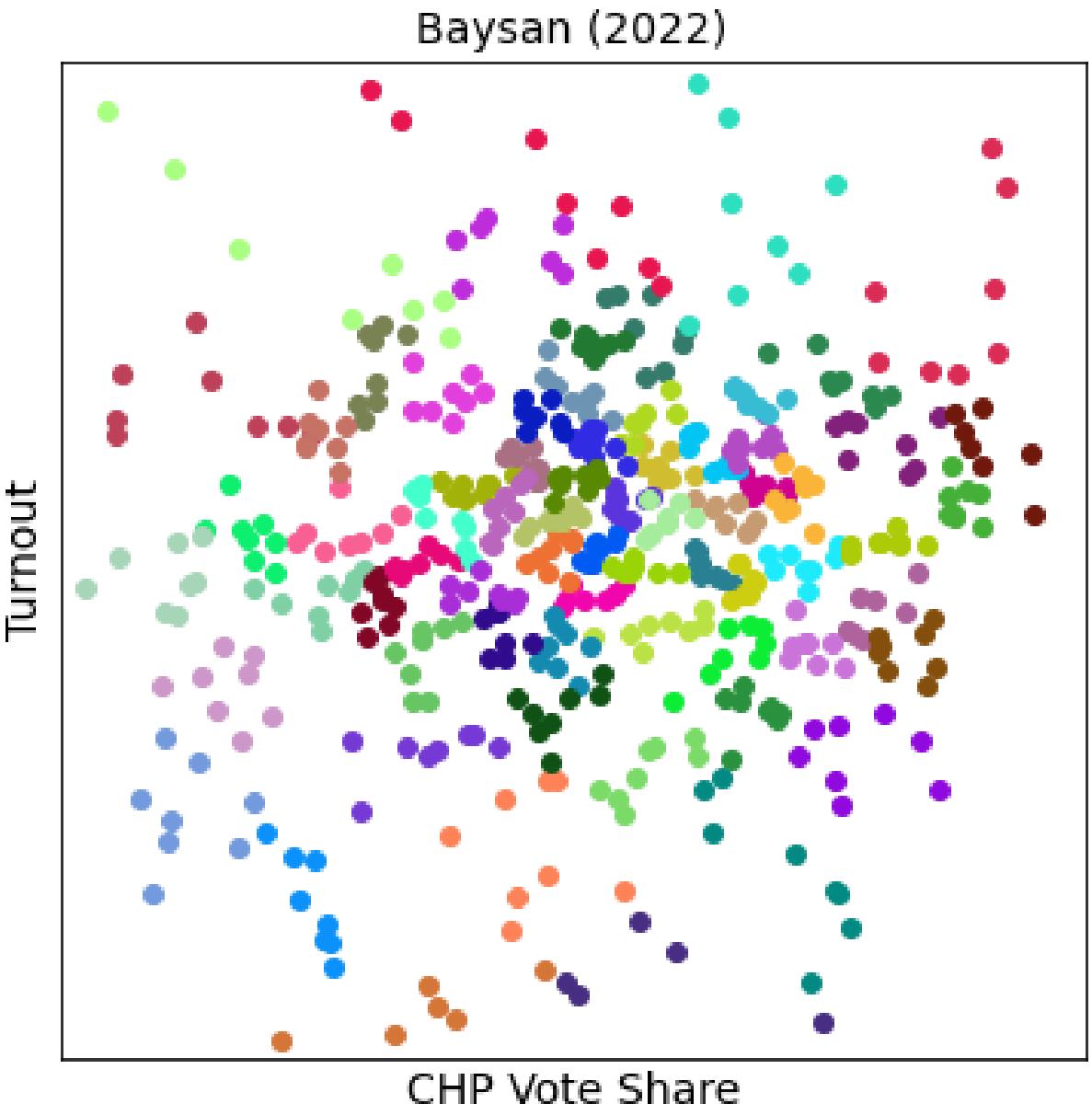
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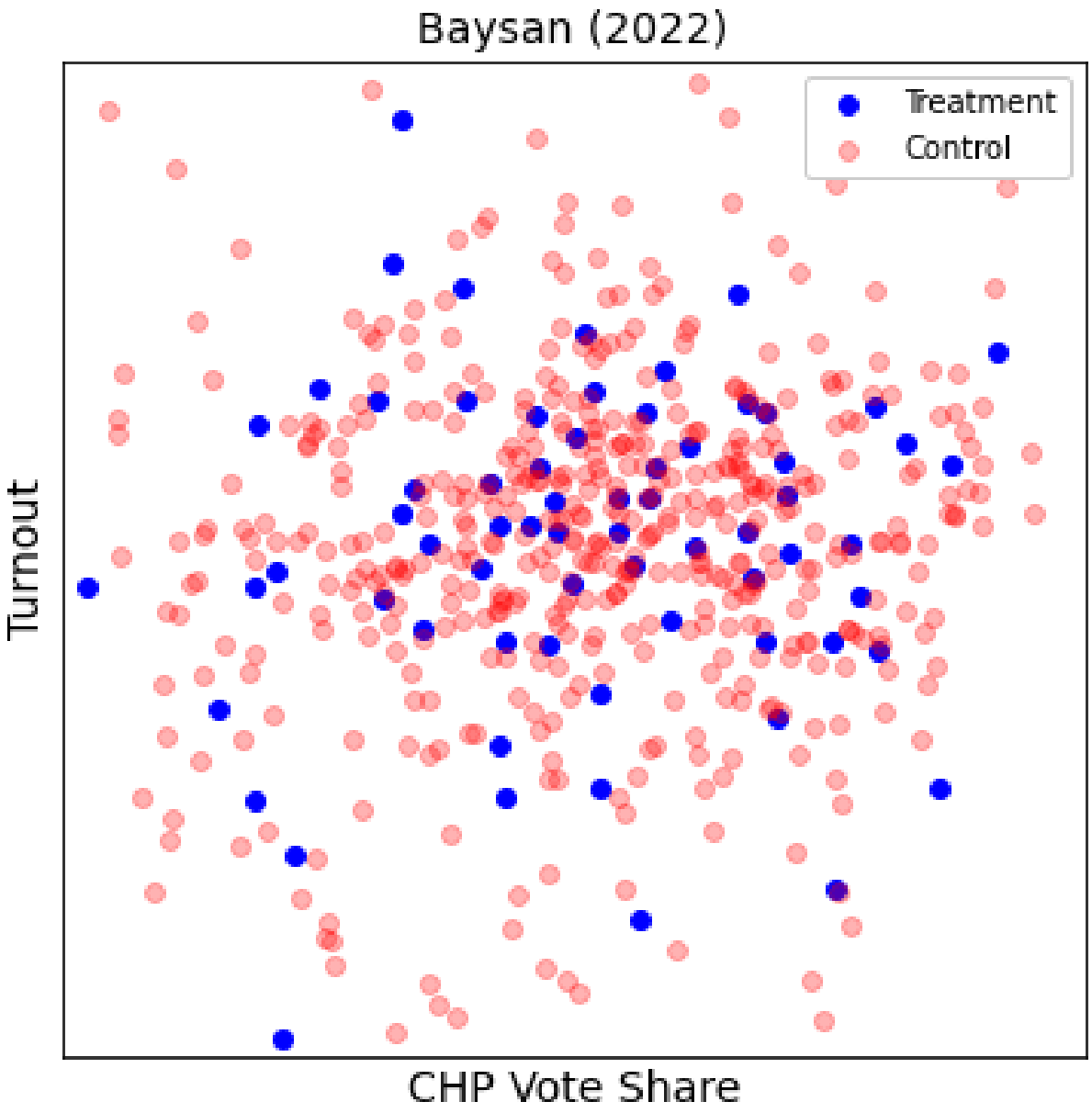
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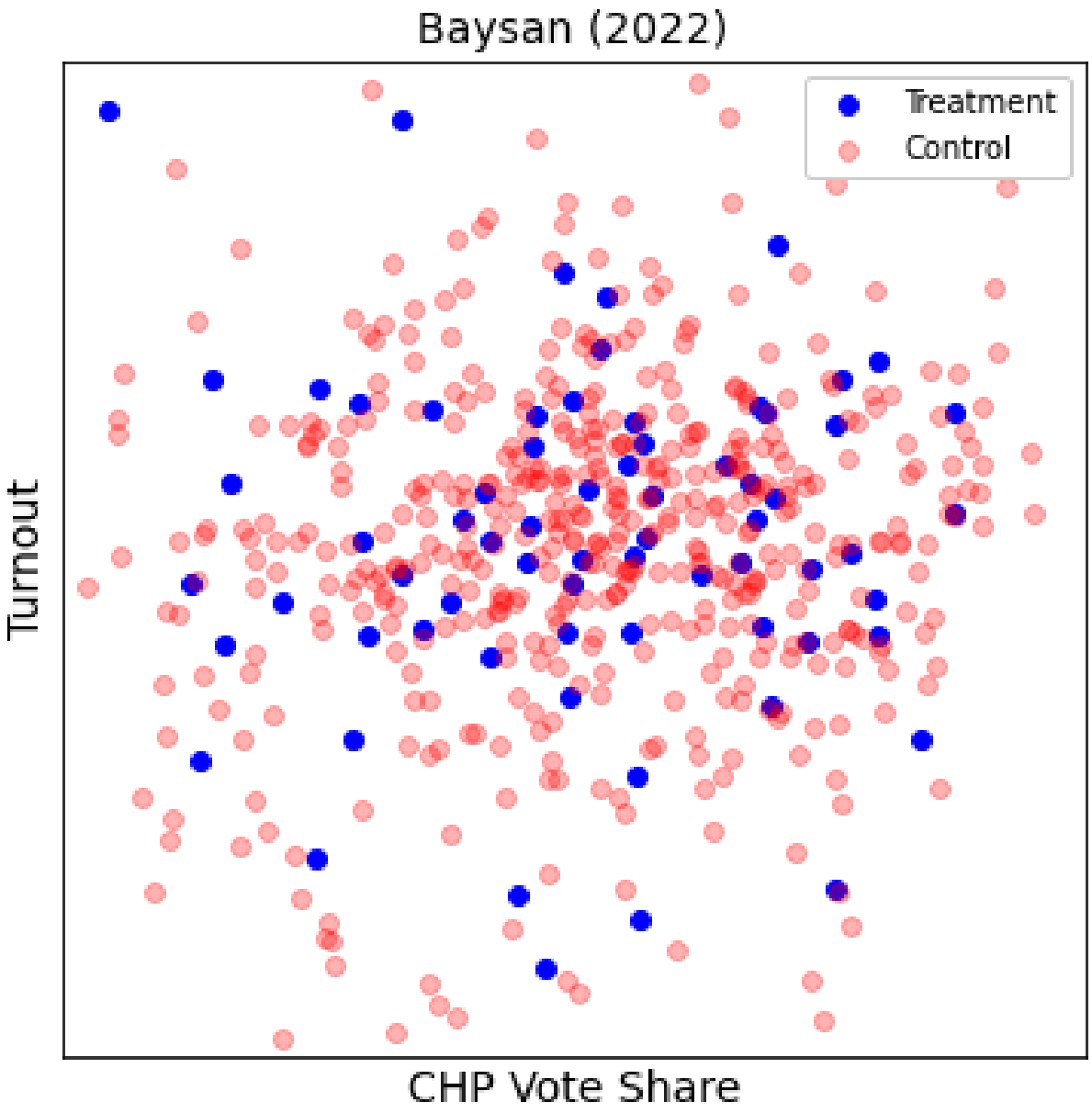
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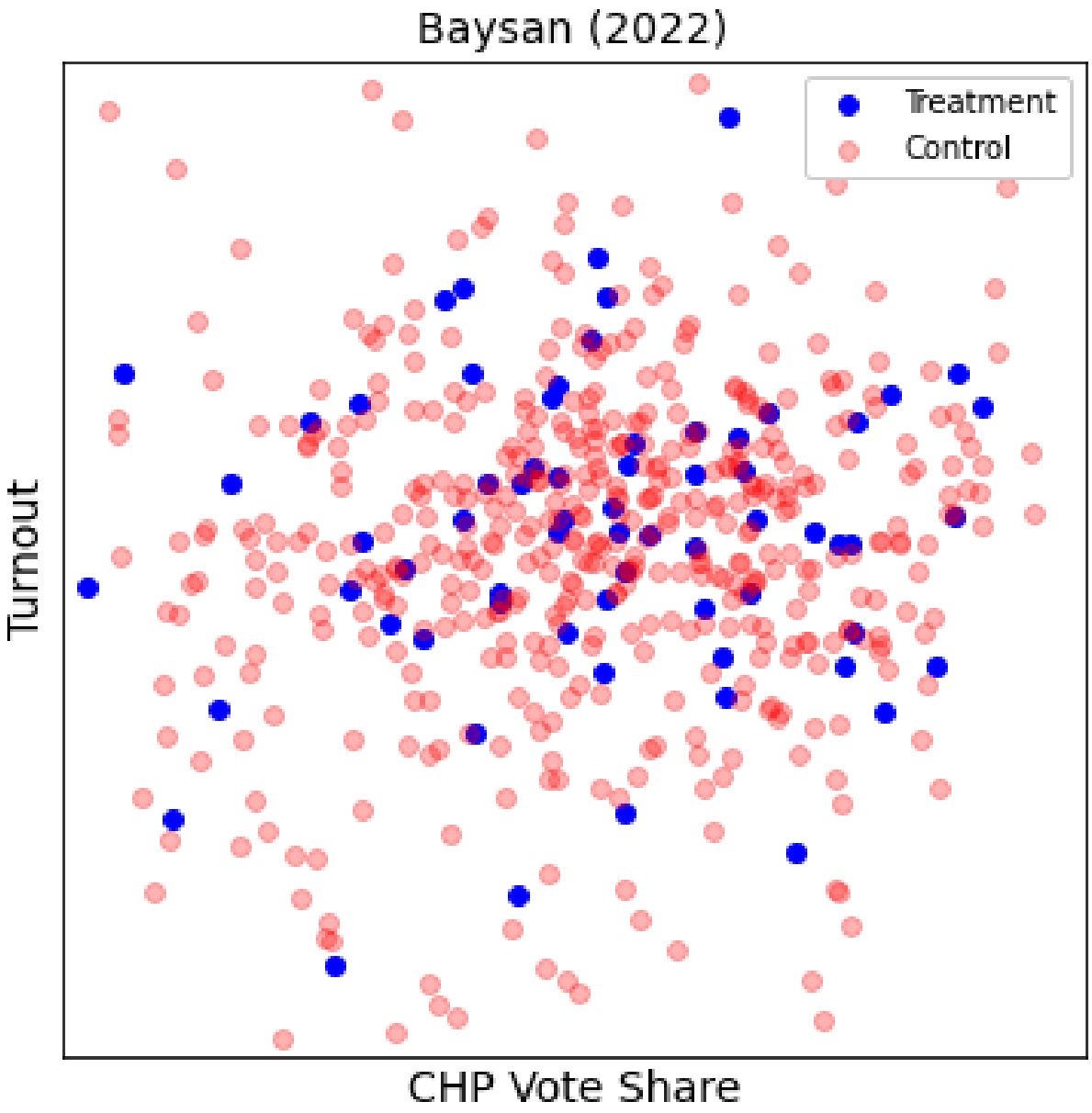
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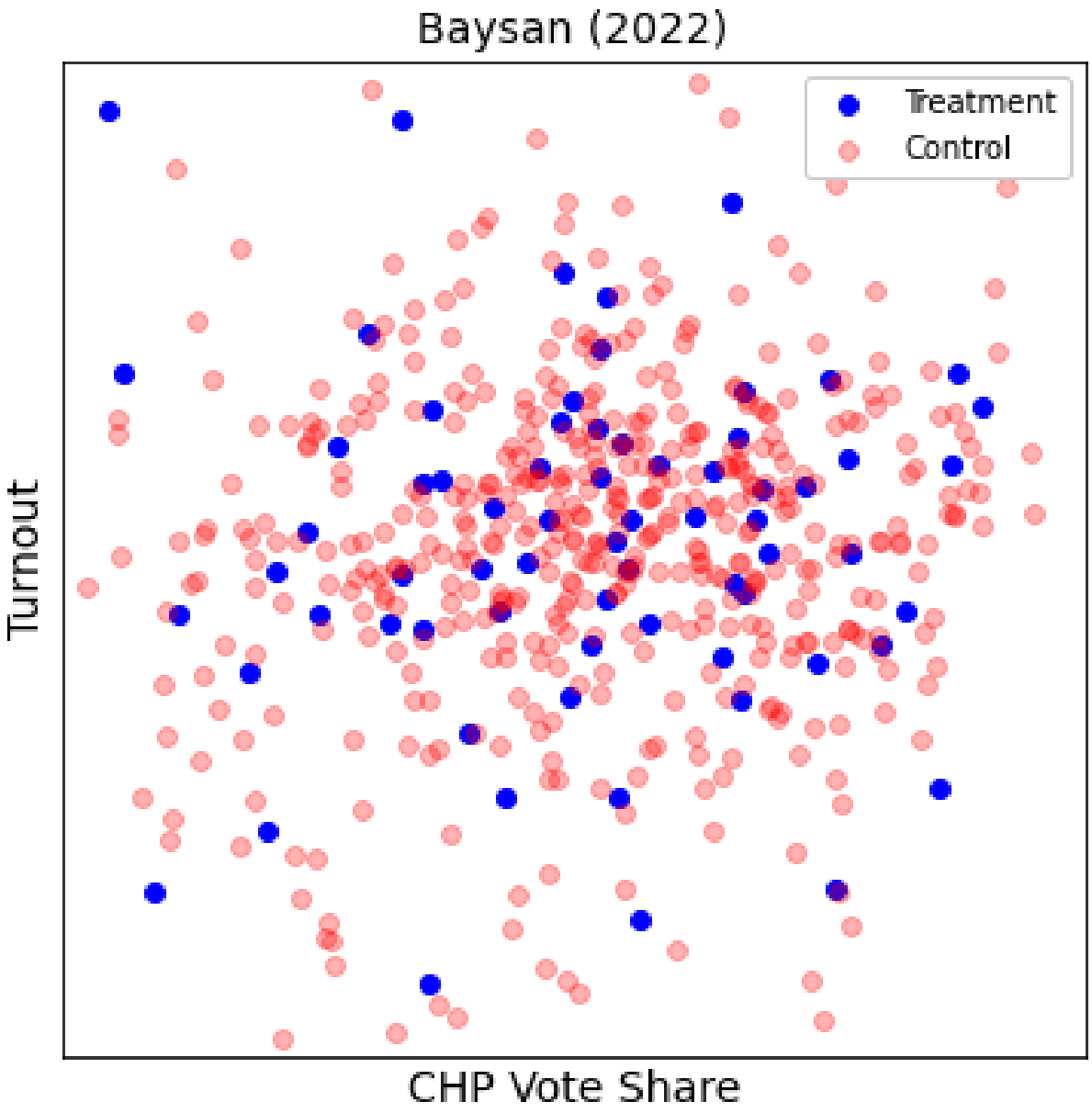
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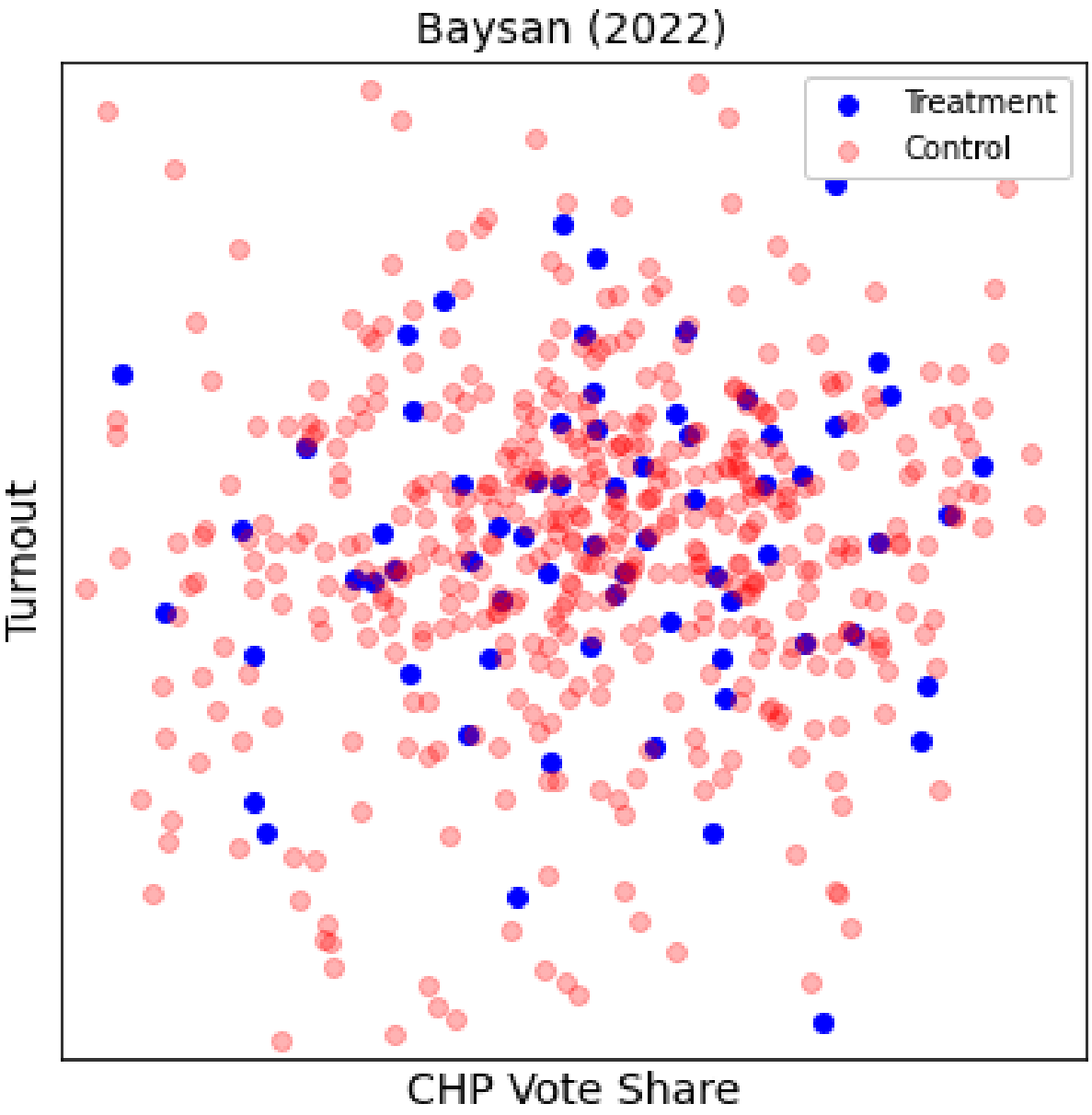
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Goal. Finely stratify to balance small set of important variables ψ nonparametrically. Rerandomize to balance linear functions of remaining variables h.

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Estimand. First, consider estimating SATE $= E_n[Y_i(1) - Y_i(0)]$, the sample average treatment effect using $\widehat{\theta}$ from $Y \sim 1 + D$.

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Warmup. If $D_{1:n} \sim \text{C}(p)$ then $\sqrt{n}(\widehat{\theta} - \text{SATE}) \Rightarrow \mathcal{N}(0, V)$,

$$V = \frac{\text{Var}(\bar{Y})}{\text{Var}(D)}.$$

Treatment Effect Estimation

Theorem. (Pure Stratification) Let $D_{1:n}$ stratified on ψ as above. Then $\sqrt{n}(\widehat{\theta} - \text{SATE}) \Rightarrow \mathcal{N}(0, V)$,

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First term reflects fluctuations of SATE $-$ ATE. Can reduce this by finely stratified sampling, e.g. see Cytrynbaum (2024).

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Theorem. Let $D_{1:n}$ assigned by stratified rerandomization as above. Then $\sqrt{n}(\widehat{\theta} - \text{SATE}) | W_{1:n} \Rightarrow \mathcal{N}(0, V^*) + R_A$, indep. RV's

$$V^* = \min_{\gamma \in \mathbb{R}^{d_h}} \frac{E[\text{Var}(\bar{Y} - \gamma' h | \psi)]}{\text{Var}(D)} = \min_{\substack{\gamma \in \mathbb{R}^{d_h} \\ t \in L_2(\psi)}} \frac{\text{Var}(\bar{Y} - \gamma' h - t(\psi))}{\text{Var}(D)}.$$

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For $\gamma_0 = \text{argmin}_{\gamma \in \mathbb{R}^{d_h}} E[\text{Var}(\bar{Y} - \gamma' h | \psi)]$, have $R_A = \gamma_0' Z_{hA}$ with

$$Z_{hA} = Z_h | Z_h \in A, \qquad Z_h \sim \mathcal{N}(0, \Sigma)$$

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Choose ϵ for acceptance probability α. $\widehat{\Sigma} \xrightarrow{p} \Sigma$, randomize until

$$\|\widehat{\Sigma}^{-1/2}\sqrt{n}(\bar{h}_1 - \bar{h}_0)\|^2 \leq \epsilon(\alpha) \quad \text{for} \quad P(\chi^2_{\dim(h)} \leq \epsilon(\alpha)) = \alpha.$$

Then number of randomizations $\approx 1/\alpha$.

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Let $W = (Y(0), Y(1), X)$. Define finite population version θ_n

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Example. (ATE) Let $g(D, Y, \theta) = HY - \theta$ for $H_i = \frac{D_i - p}{p - p^2}$. Then we have $\theta_0 = \text{ATE}$ and $\theta_n = \text{SATE}$.

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$f(X, \theta) = X'\theta$ gives BLP of LATE. $f(X, \theta) = \theta$ recovers LATE.

General Causal Parameters

Finite Population. $\sqrt{n}(\widehat{\theta}_{GMM} - \theta_n)|W_{1:n} \Rightarrow \mathcal{N}(0, V^*) + R_A$

$$V^* = \min_{\substack{\gamma \in \mathbb{R}^{d_h \times d_\theta} \\ t \in L_2(\psi)^m}} \mathrm{Var}\left(\phi_a - \gamma' h - t(\psi)\right) / \mathrm{Var}(D).$$

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Superpopulation. $\sqrt{n}(\widehat{\theta}_{GMM} - \theta_0) \Rightarrow \mathcal{N}(0, \mathrm{Var}(\phi_s) + V^*) + R_A$ for a "sampling" influence function ϕ_s.

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E.g. for $\theta_0 = \mathrm{argmin}_\theta E[(Y(1) - Y(0) - X'\theta)^2]$, have $\phi_a = E[XX']^{-1}\bar{Y}X$. Choose ψ, h predictive of $\bar{Y}X$.

Nonlinear Rerandomization

Are we missing something by using $\sqrt{n}(\bar{h}_1 - \bar{h}_0) \in A$? Want to rerandomize until distributions $h_i | D_i = 1 \approx h_i | D_i = 0$.

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GMM Rerandomization. Stratify on ψ, draw $D_{1:n}$ and estimate

$$E_n[D_i m(X_i, \widehat{\beta}_1)] = 0, \qquad E_n[(1 - D_i) m(X_i, \widehat{\beta}_0)] = 0.$$

Rerandomize until $\sqrt{n}(\widehat{\beta}_1 - \widehat{\beta}_0) \in A$.

GMM Rerandomization

$$m(X, \beta) = X - \beta \qquad \text{(linear)}$$
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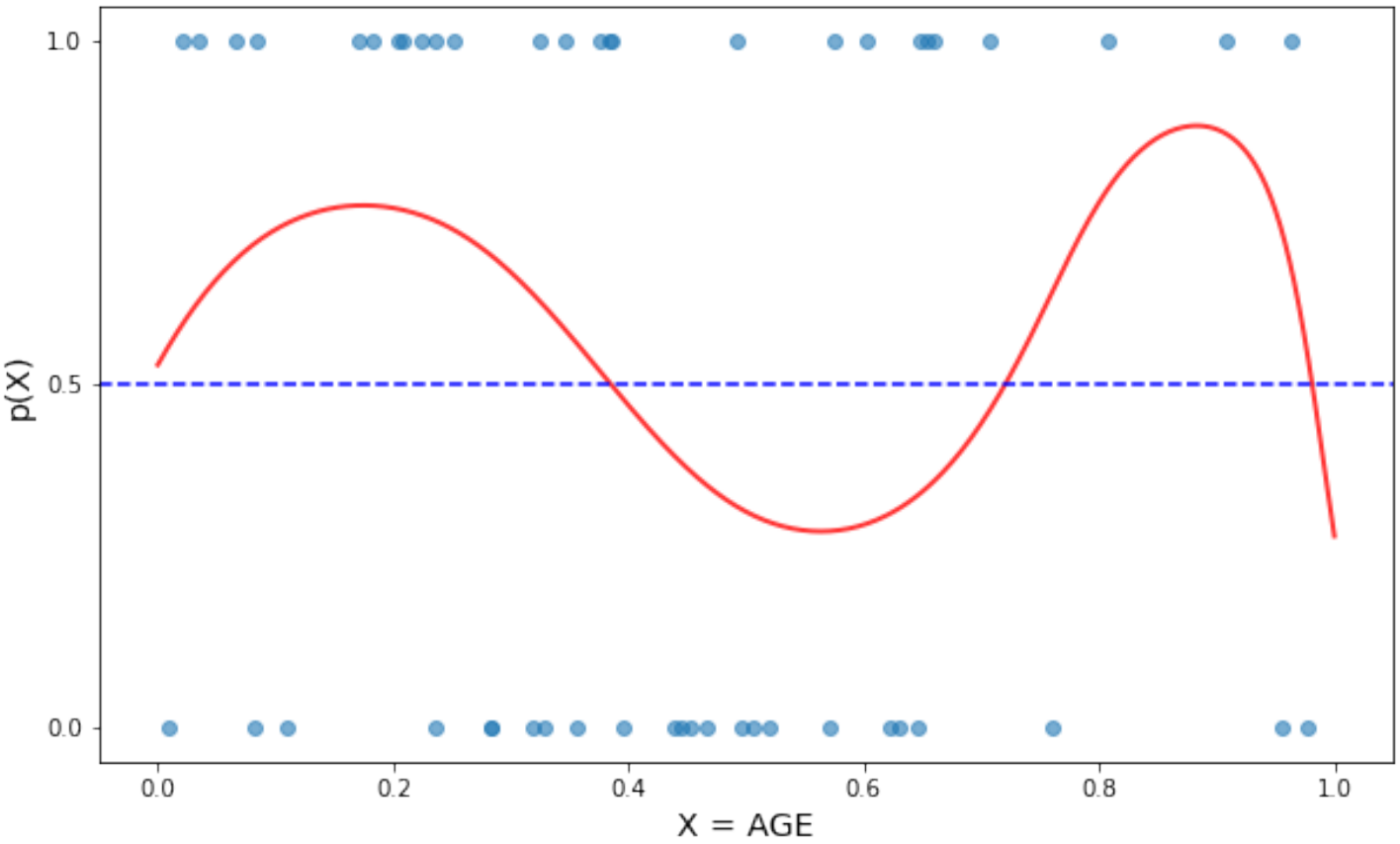
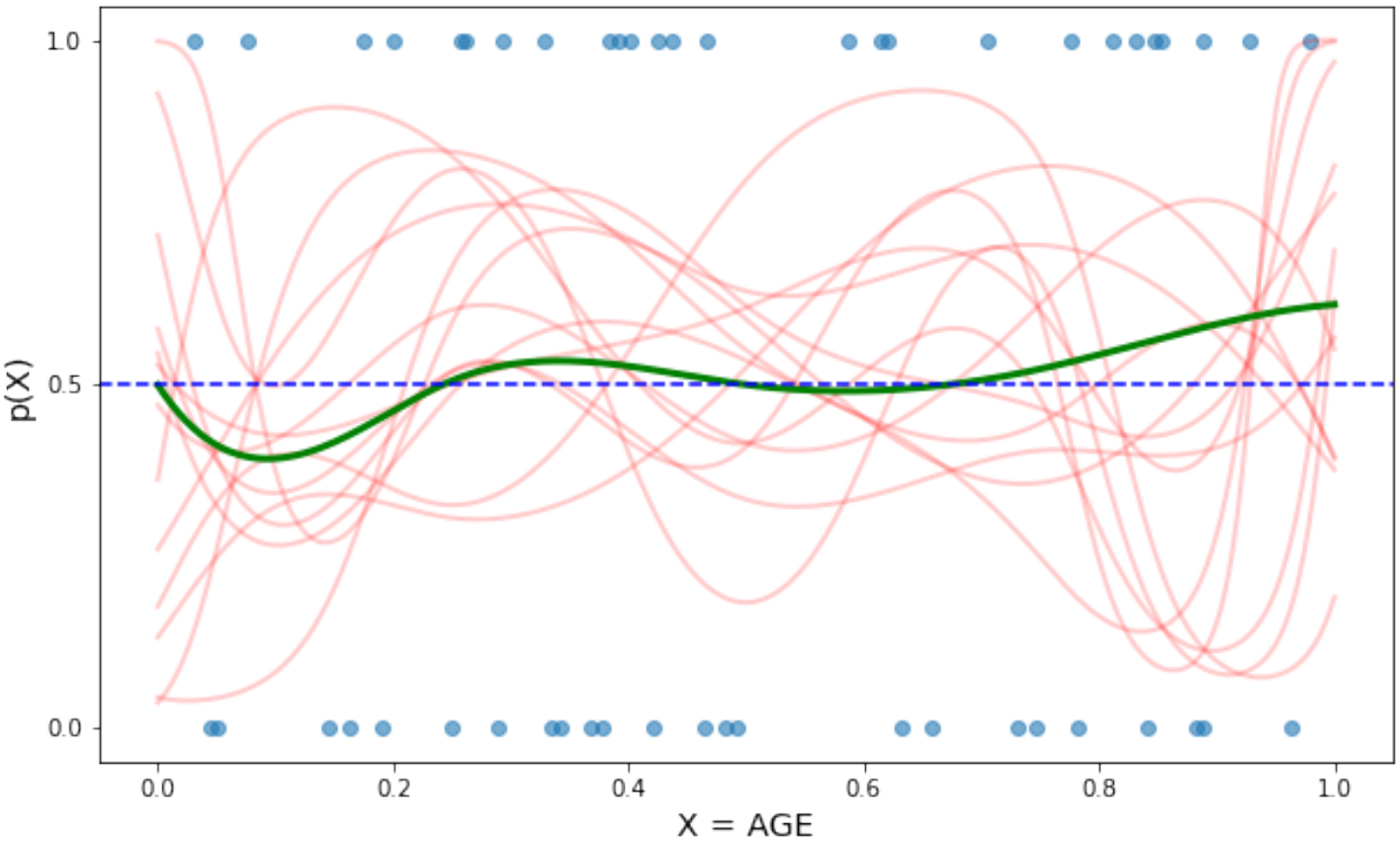
Propensity Rerandomization. Rerandomize until X has no predictive power for D according to the propensity $\widehat{p}(X)$.


1.0
0.5
0.0
p(X)
0.0
0.2
0.4
0.6
0.8
1.0
X = AGE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Result. General rerandomization criteria based on GMM or propensity score estimation equivalent to linear rerandomization with implicit choices of h and A.

Optimizing Acceptance Regions

$$\sqrt{n}(\widehat{\theta} - \text{SATE}) \Rightarrow \mathcal{N}(0, V^*) + \gamma_0' Z_{hA}$$

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Infeasible since γ_0 is unknown at design-time.

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Minimax. Consider prior belief set $B \subseteq \mathbb{R}^{d_h}$. Rerandomize until:

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Example. For guess $\gamma_0 \approx \bar{\gamma}$ and uncertainty δ, set $B = B_2(\bar{\gamma}, \delta)$,

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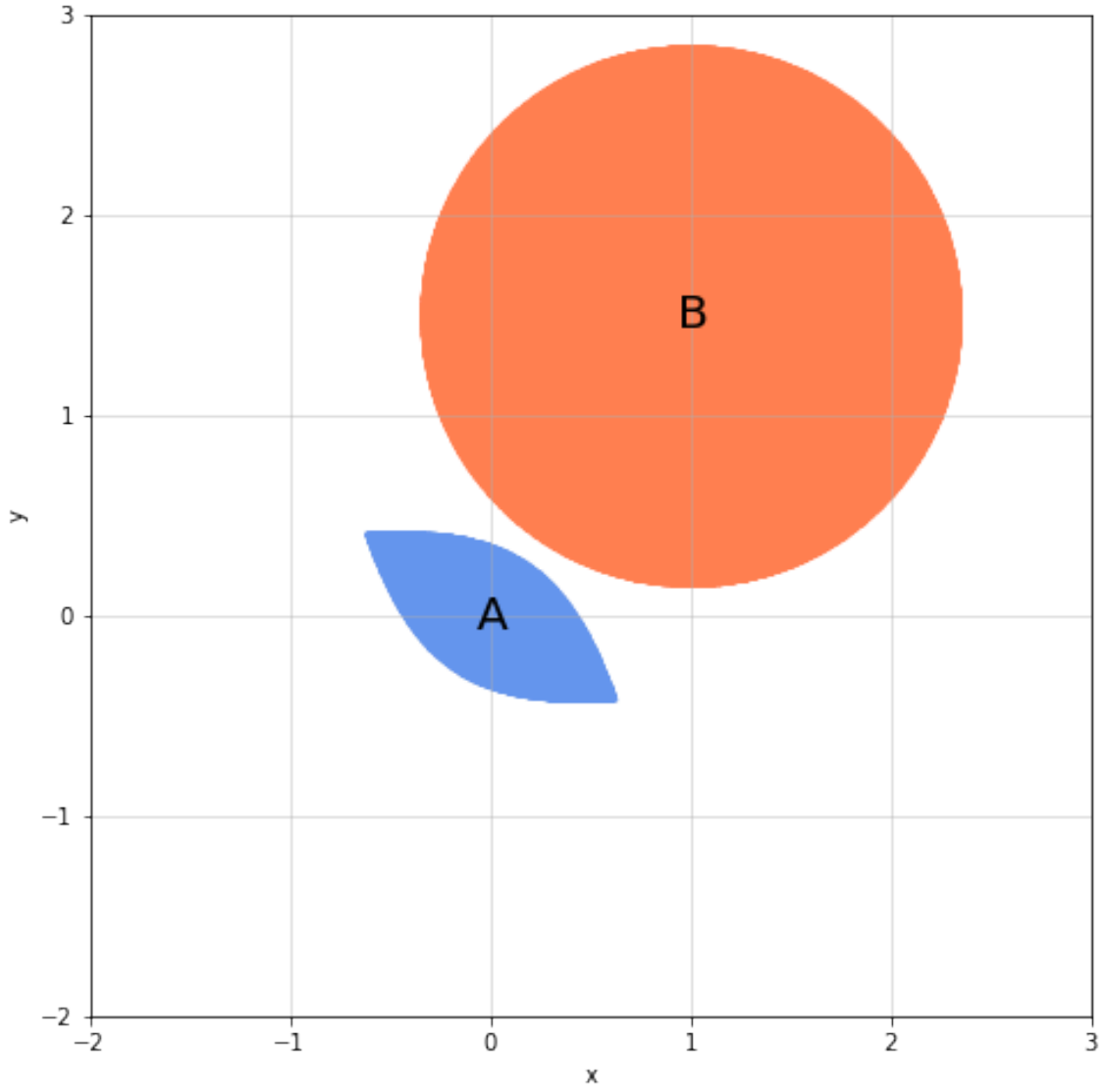
B
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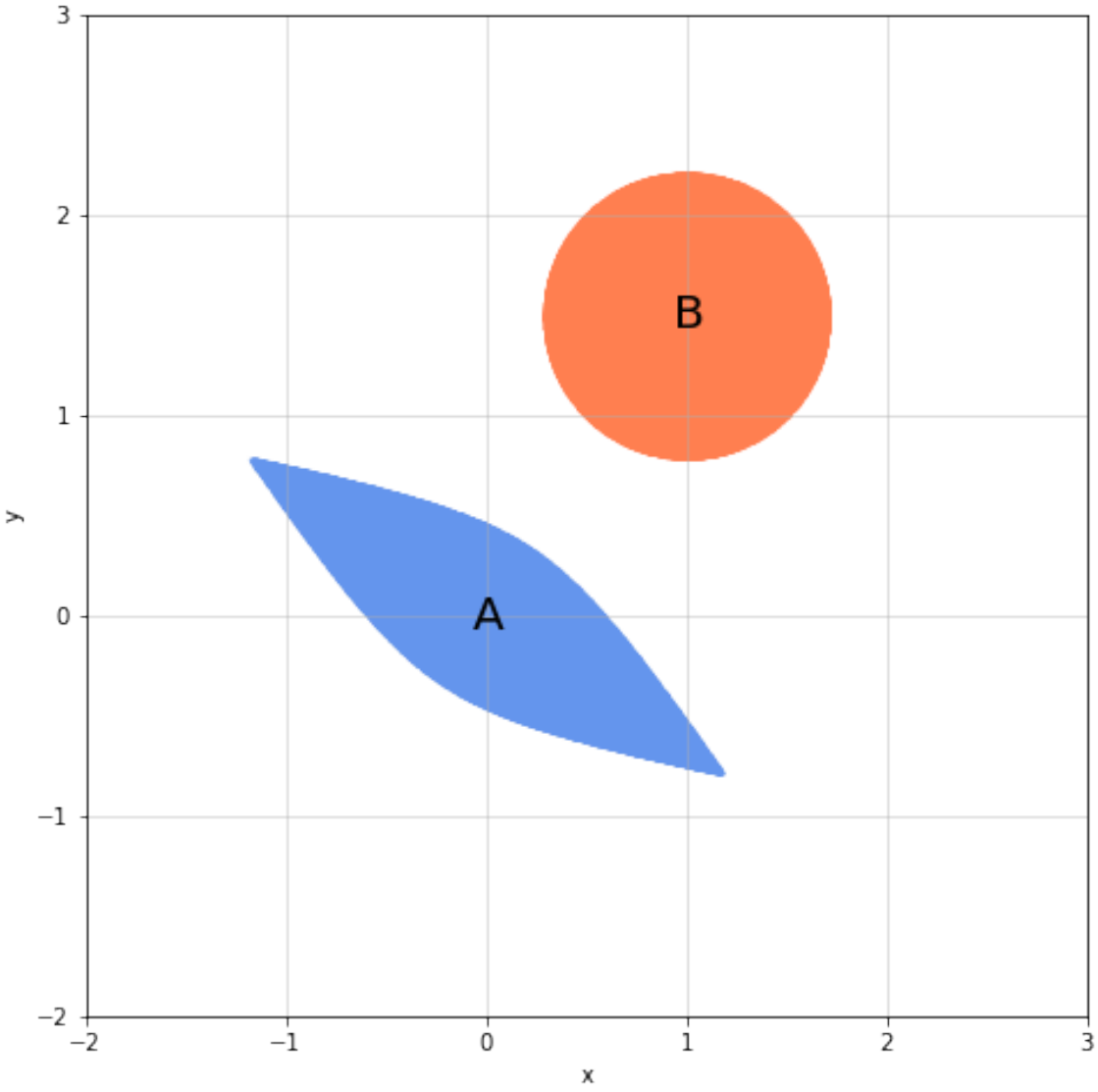
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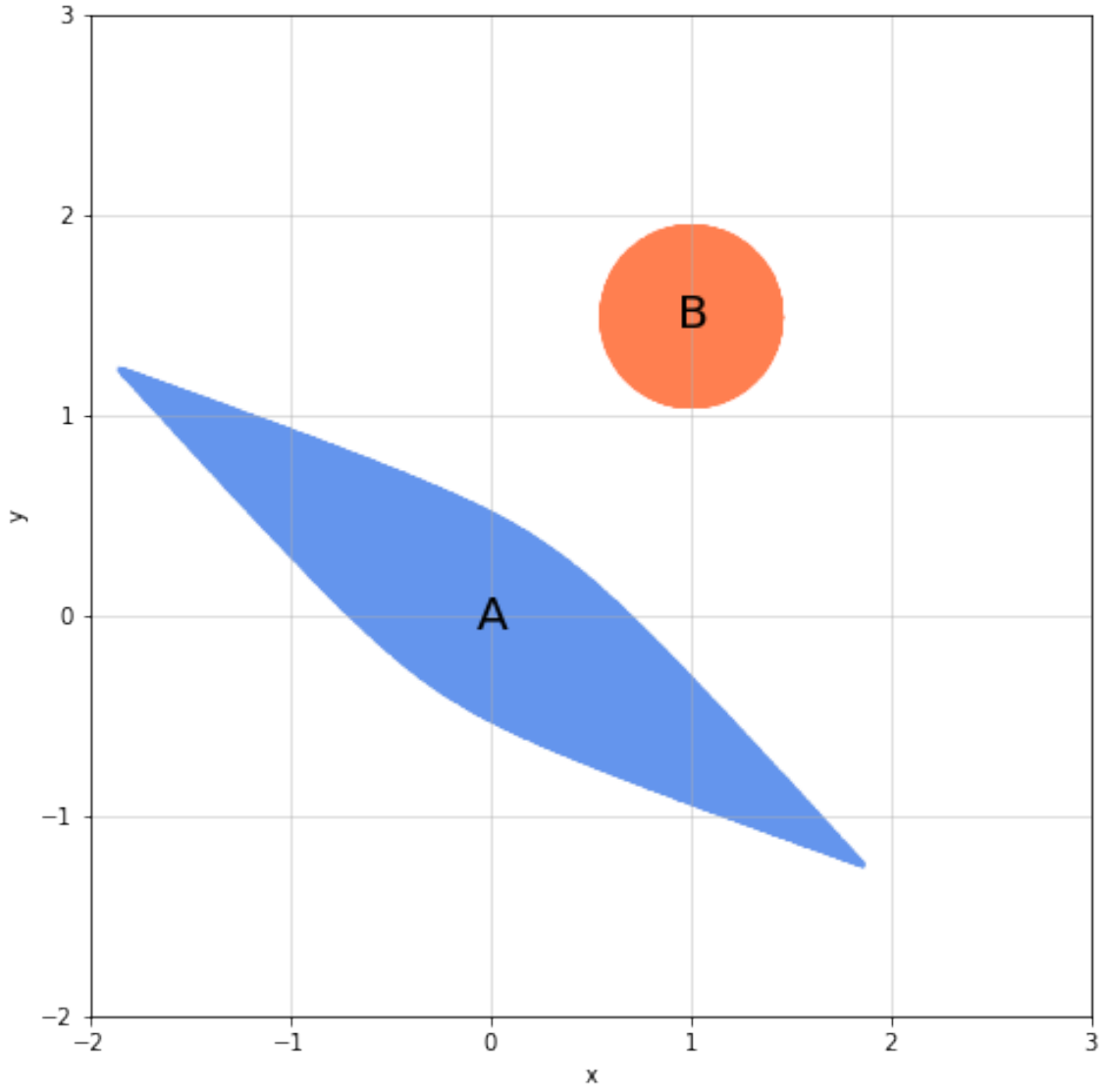
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Intuitively, use prior information about outcome model to choose shape of A, form of dimension reduction.

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For larger $\dim(h)$, combining rerandomization and adjustment provides meaningful finite sample efficiency gains, due to a novel double robustness property.

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Double Robustness. For $\widehat{\theta}_{adj} = \widehat{\theta} - \widehat{\gamma}'(\bar{h}_1 - \bar{h}_0)$, have

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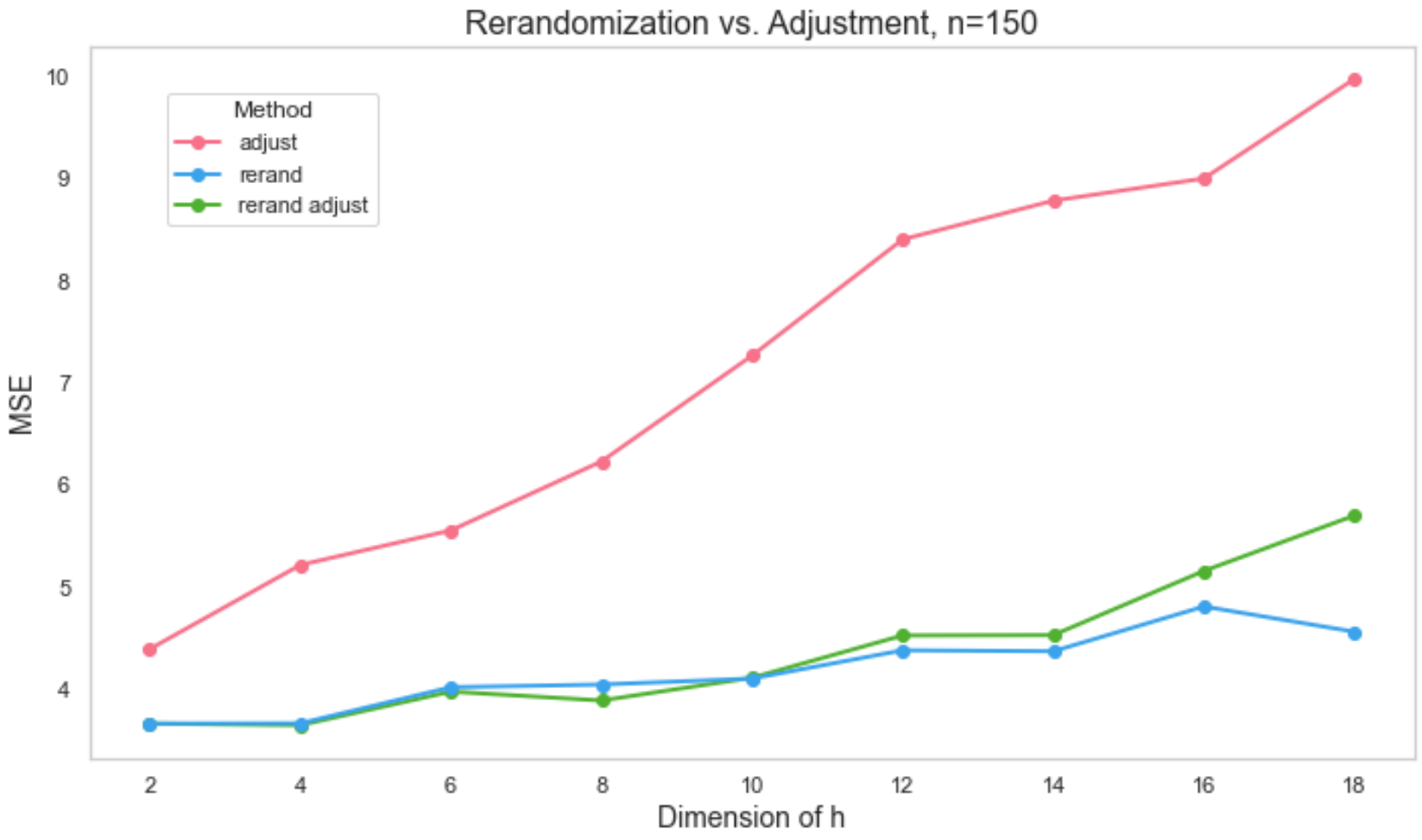
Efficiency


Rerandomization vs. Adjustment, n=150
Method
adjust
rerand
rerand adjust
MSE
10
9
8
7
6
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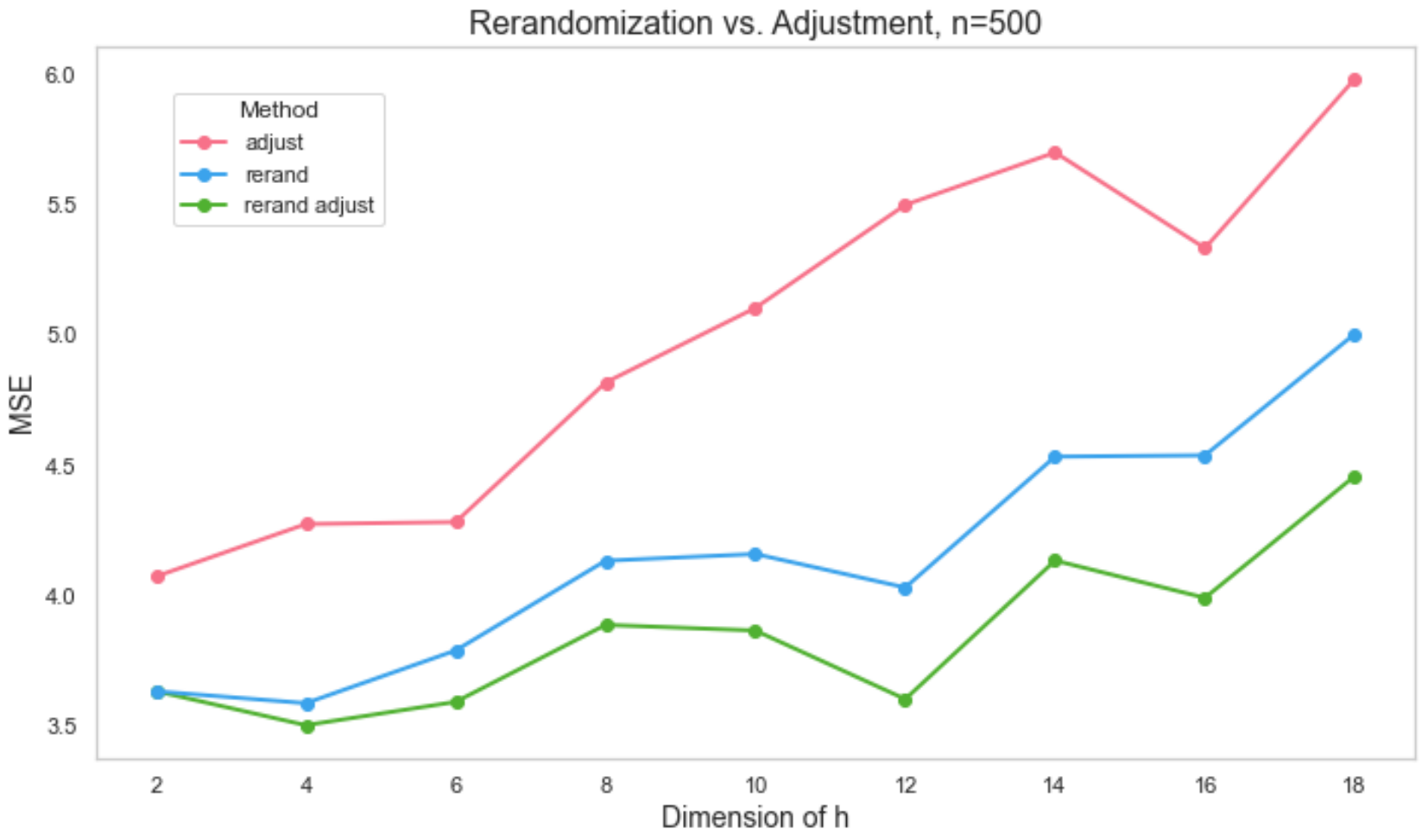
4
2
4
6
8
10
12
14
16
18
Dimension of h

Efficiency


Rerandomization vs. Adjustment, n=500
Method
adjust
rerand
rerand adjust
MSE
6.0
5.5
5.0
4.5
4.0
3.5
2
4
6
8
10
12
14
16
18
Dimension of h

Inference

For $\widehat{\theta}_{adj} = \widehat{\theta}_{GMM} - \widehat{\gamma}'(\bar{h}_1 - \bar{h}_0)$, recover $\sqrt{n}(\widehat{\theta}_{adj} - \theta_n) \Rightarrow \mathcal{N}(0, V^*)$.

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The variance V^* is not identified. For $\theta_n = \text{SATE}$, $D_{1:n} \sim \text{C}(p)$

$$V^* = \frac{\text{Var}(\bar{Y})}{\text{Var}(D)} \propto \text{Cov}(Y(1), Y(0))$$

Inference

For $\widehat{\theta}_{adj} = \widehat{\theta}_{GMM} - \widehat{\gamma}'(\bar{h}_1 - \bar{h}_0)$, recover $\sqrt{n}(\widehat{\theta}_{adj} - \theta_n) \Rightarrow \mathcal{N}(0, V^*)$.

The variance V^* is not identified. For $\theta_n = \text{SATE}$, $D_{1:n} \sim \text{C}(p)$

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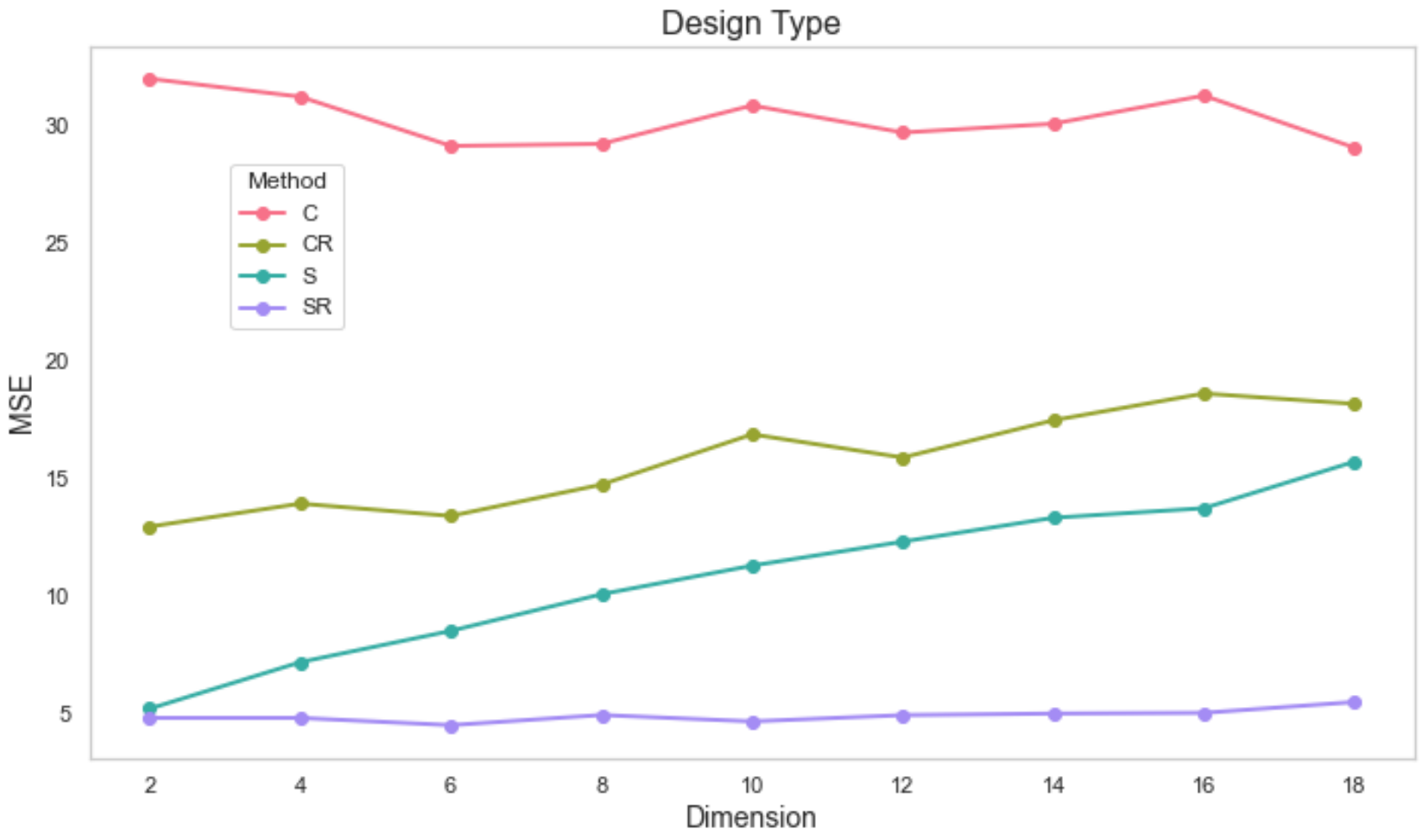
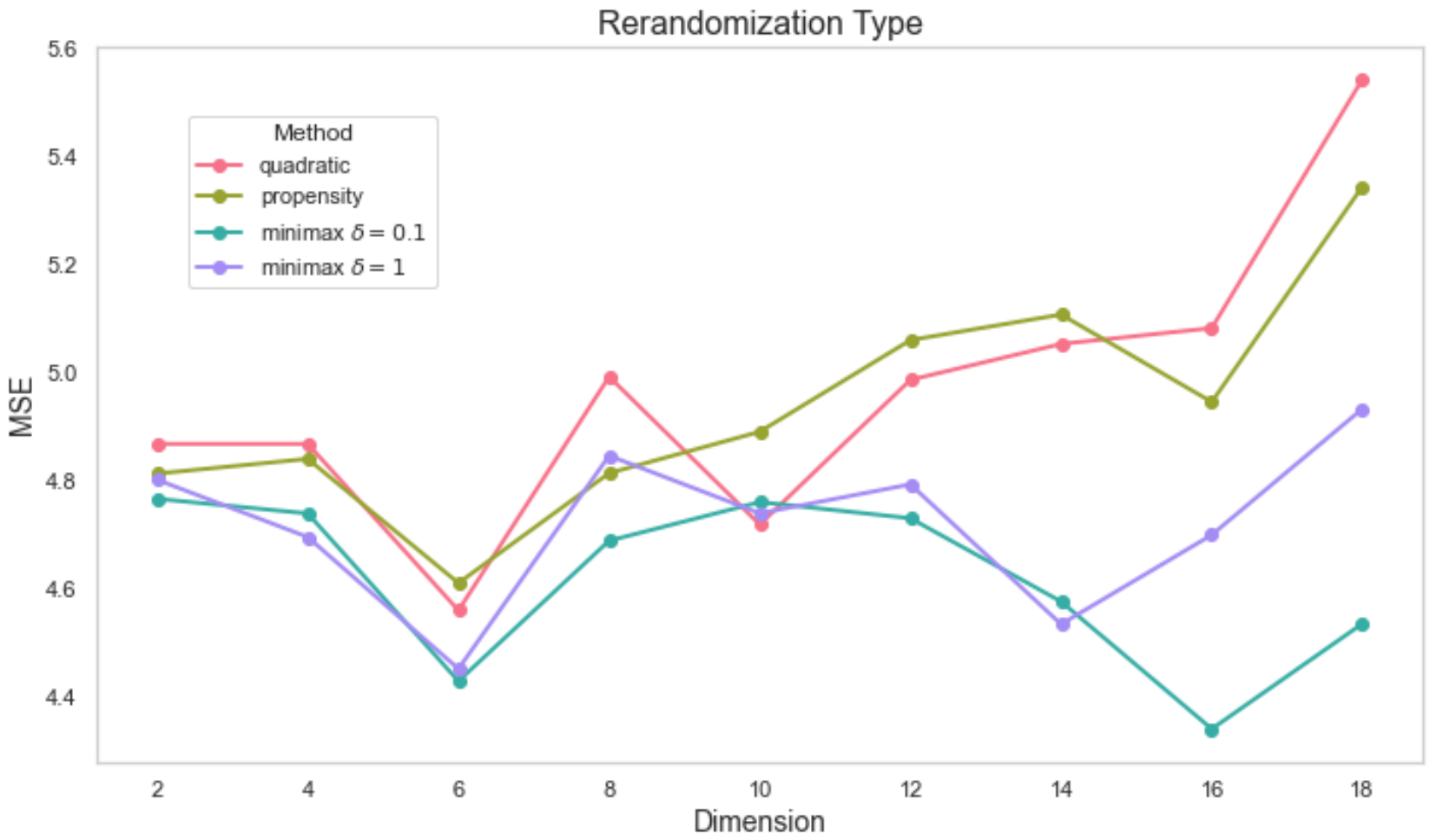
Superpopulation. Asymptotically exact inference on θ_0. Both use "collapsed strata" type methods to deal with small strata.

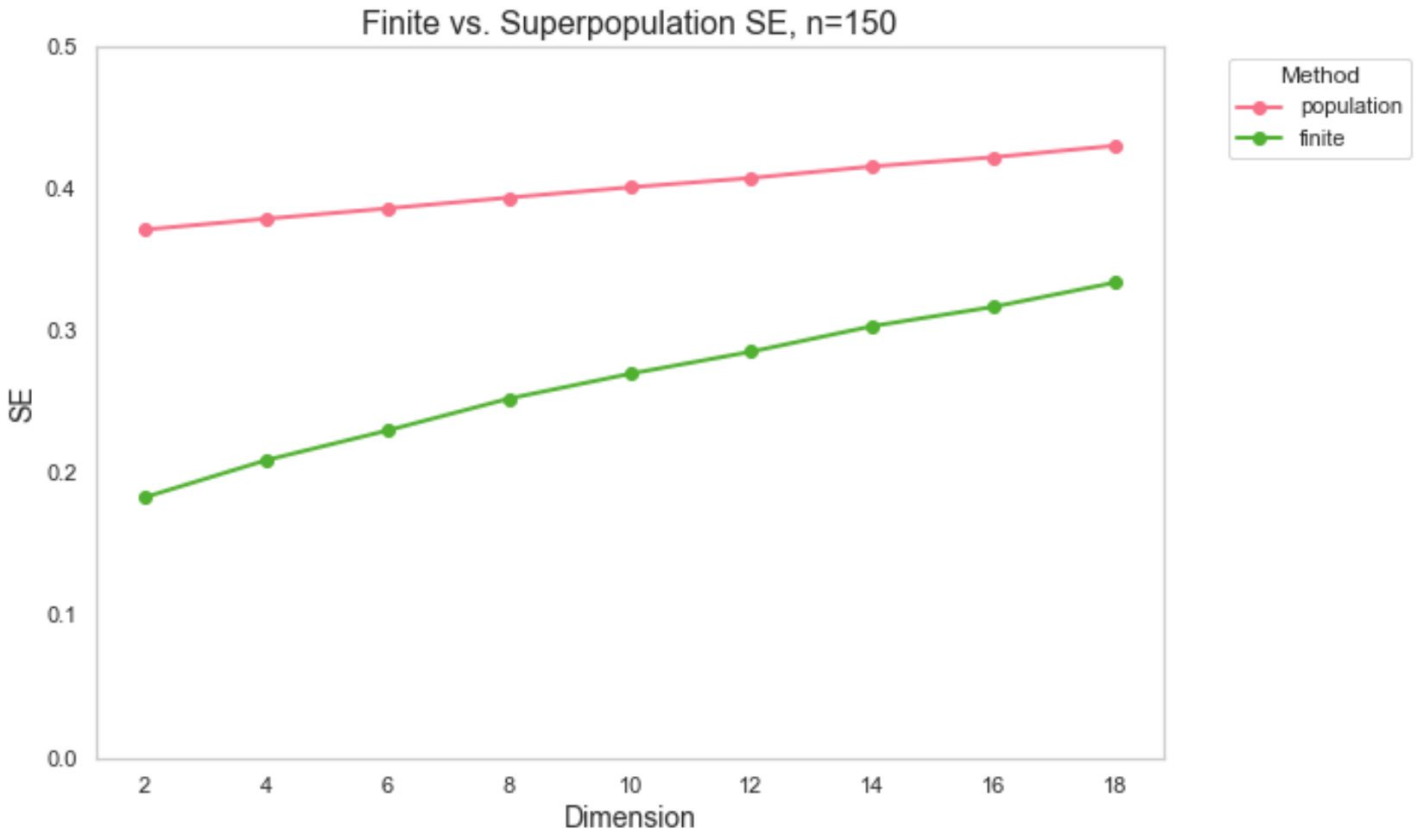
Efficiency


Design Type
Method
C
CR
S
SR
MSE
30
25
20
15
10
5
2
4
6
8
10
12
14
16
18
Dimension

Efficiency


Rerandomization Type
Method
quadratic
propensity
minimax $\delta = 0.1$
minimax $\delta = 1$
MSE
5.6
5.4
5.2
5.0
4.8
4.6
4.4
2
4
6
8
10
12
14
16
18
Dimension

Inference


Finite vs. Superpopulation SE, n=150
0.5
0.4
0.3
0.2
0.1
0.0
SE
2
4
6
8
10
12
14
16
18
Dimension
Method
population
finite

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Angrist et al. (2013), "When Opportunity Knocks, Who Answers?".

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Estimate LATE and BLP of LATE

$$\min_{\theta} E_n[(Y_i(1) - Y_i(0) - X_i'\theta)^2 | D_i(1) > D_i(0)]$$

Efficiency for BLP of LATE

		MSE		Cover		CI Width	
θ_n (LATE)	Design	$\widehat{\theta}$	$\widehat{\theta}_{adj}$	Pop.	Fin.	Pop.	Fin.
CLATE (GPA)	C	3.06	1.00	0.92	0.98	1.00	1.02
	CR	1.76	0.85	0.95	0.99	0.97	0.97
	S	1.42	0.98	0.94	0.98	0.97	1.01
	SR	1.07	0.89	0.94	0.99	0.97	0.99
	F	0.86	0.92	0.97	0.98	1.10	0.97
	FR	0.79	0.83	0.97	0.99	1.05	0.94
	F+	1.41	1.44	0.96	0.95	1.39	1.32
	FR+	1.32	1.34	0.96	0.97	1.38	1.32

Table: LATE Parameters

Summary

Stratified rerandomization does partially linear regression adjustment "by design."

Results hold for generic causal parameters in a GMM framework.

"Nonlinear" rerandomization designs (e.g. density-based) asymptotically equivalent to linear rerandomization.

Minimax scheme optimizing shape of the acceptance region.

Optimal ex-post linear adjustment restores asymptotic normality, enabling simple and efficient inference methods.